

CS 122A Custom Laboratory Project Report

The Allan Triggerer 3000

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Introduction

This section should describe the project that was implemented, encompassing basic functionality. Similar to the project proposal, this section should not describe technical/implementation aspects.

This section should also state any important aspects of the project that were not successfully completed, including any major bugs/errors that limit usability.

The Allan Triggerer 3000 is a real-time face detection system that plays country music whenever it detects a human face. This was specifically designed to torment Allan, who hates country music. Using a camera module, the system continuously monitors its field of view and triggers an audio playback response upon detecting a face.

Elements of Complexity

Explain the parts of your project that you consider to be elements of complexity and what you think makes your project stand out. Clearly distinguish between elements of complexity that were successfully implemented, those that were not fully implemented, or those that were not implemented at all.

User Guide

This system is designed to operate with minimal user interaction. The primary input is a camera that continuously monitors its field of view for the presence of a human face.

Under normal operating conditions, the camera captures live video and the face detection software analyzes each frame. When a human face is detected for the first time, the system automatically triggers audio playback and begins playing a predefined country music song through a connected speaker.

Once the song begins playing, it will continue uninterrupted until the end of the competition. The audio playback does not stop if the detected face leaves the camera's field of view, becomes obscured, or is no longer detected. Face detection is only used to initiate the song playback.

Inputs:

- Live video feed from the camera
- Presence of a human face within the camera's field of view

Outputs:

- Audio playback of a country music song through a speaker

Normal Operation Sequence:

1. The system powers on and initializes the camera and audio components.
2. The camera continuously scans for a human face.
3. When a face is detected, the system starts playing the country song.
4. The song continues playing for the remainder of the competition regardless of whether the face remains visible.
5. No further user interaction is required after the song has been triggered.

Hardware Components Used

List all of the hardware components that were used in the project. A simple bullet-point list is fine; a detailed description of the components is unnecessary.

- IceSugarPro FPGA
- Raspberry Pi Pico W
- OV2640 Camera Module
- DFPlayer Mini
- Gikfun Speaker
- RGB LCD
- 2N2222A
- Potentiometer

Software Libraries Used

- `pico_stdlib`
 - The Raspberry Pi Pico standard library, part of the Pico SDK. It provided core functions such as `stdio_init_all()`, `gpio_init()`, `gpio_set_dir()`, `gpio_put()`, `gpio_get()`, `sleep_ms()`, and `gpio_get_all()`, which were essential for initializing the board, reading camera data signals (VSYNC, HREF, PCLK, D0–D7), and controlling the LED output.
- `hardware_i2c`
 - The Pico SDK's hardware I2C library. It was used to communicate with the OV2640 camera module over I2C (via `i2c_init()`, `i2c_write_blocking()`, `gpio_set_function()`, and `gpio_pull_up()`), allowing the program to send

configuration register values to initialize the camera into QVGA and RGB565 modes.

Protocols Used

- I2C
 - Used to configure the OV2640 camera sensor by writing register value pairs to the camera's I2C address. The Pico acted as the I2C controller. This allowed the initialization tables to be applied to set up the camera's resolution and output format.
- Parallel Camera Interface
 - Used to receive raw pixel data from the OV2640 after configuration. The camera outputs pixel bytes in parallel across 8 data pins, synchronized by a pixel clock with row and frame timing controlled by HREF and VSYNC. The firmware bit-bangs this interface to read RGB565 pixel pairs and perform skin-tone detection frame by frame.

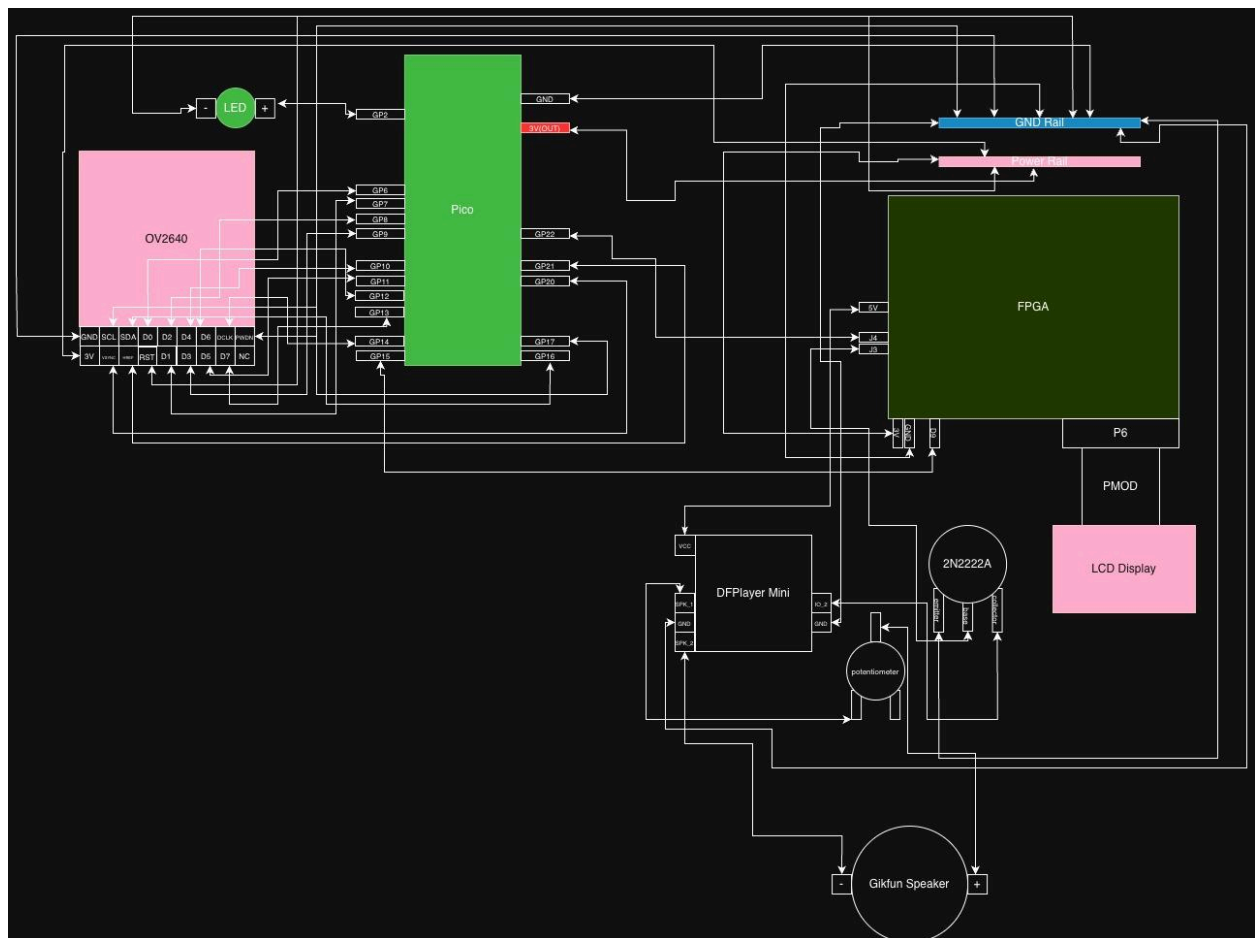
Requirements

- FPGA will be in charge of handling the audio playback
 - For the audio playback, the FPGA controlled the playback by driving the DFPlayer module through a 2N2222A transistor. When the tone detection goes off, the FPGA creates a trigger pulse on J3, which drives the transistor to trigger the DFPlayer's IO2 pin to play the next song. The DFPlayer is connected to the external speakers.
- FPGA will display a signal on whether a face is detected
 - For the signal on the LCD display, we had 2 states: a red screen for no detection and a screen that had an avatar of Allan in a cowboy outfit for detection. This was done through the PMOD adapter.
- The Pico will handle the image capturing from the camera module
 - The OV2640 Camera Module was wired through the Pico, configuring the camera over SCCB. We used the camera's 8-bit parallel data bus to capture the pixel data. The Pico reads the RGB565 pixels and runs a skin-tone detection algorithm to determine if the color percentages meet the threshold to count as a valid skin-tone detection.

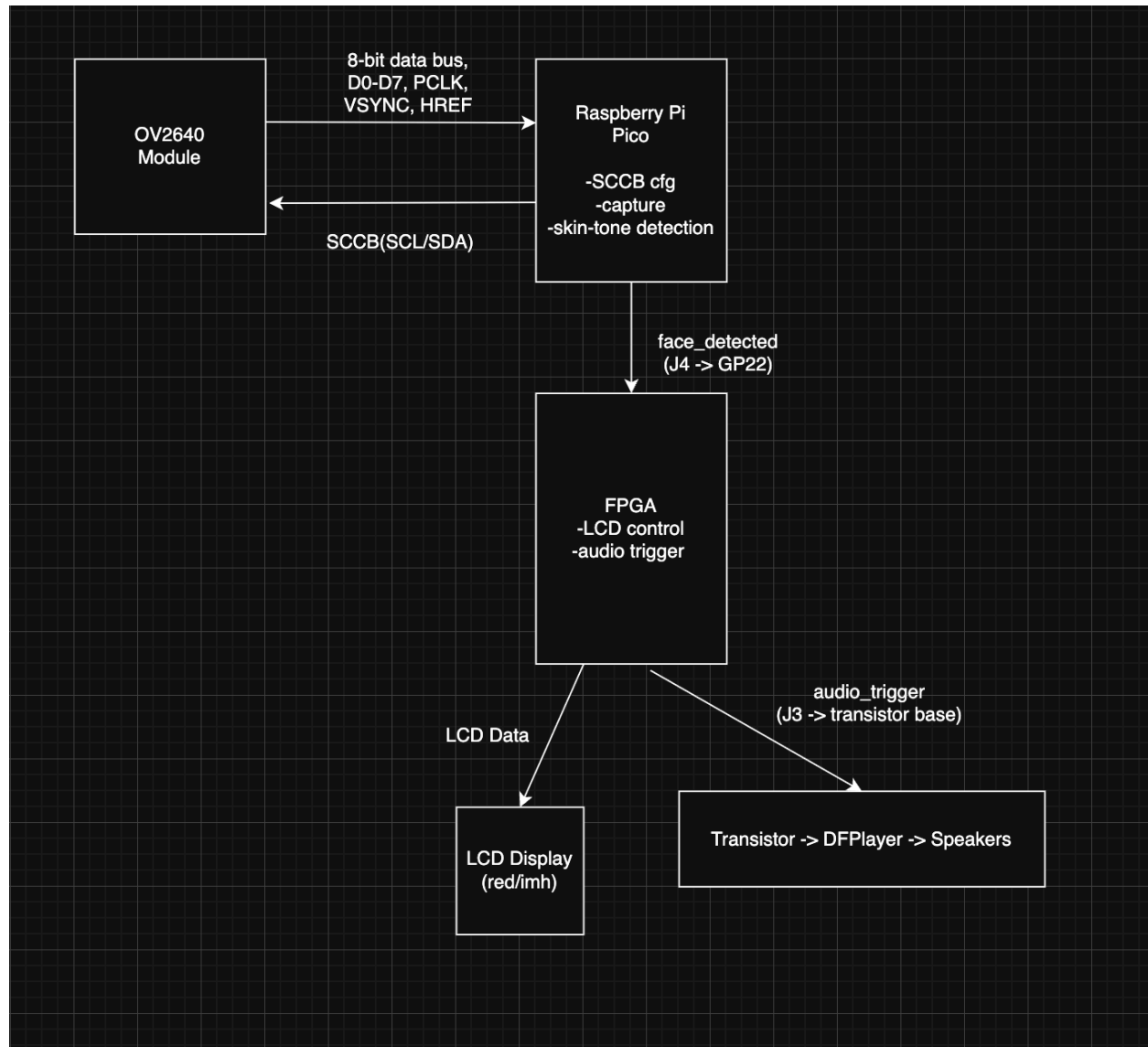
- When the FPGA is triggered by a specific event, it will send that information to the speaker system
 - The FPGA receives the face_detected signal from the Pico, updates the LCD display and emits the audio trigger pulse for the speaker.

Wiring Diagram

Take a digital photograph of your physical setup and place it next to or below the diagram.



Design Diagram



AI Usage

State what parts of your project you used AI for(if any) and what AI helped you accomplish in those parts

There was some use of AI in our project. The main use was understanding how to use the parts and what libraries and functions would allow us to achieve our project goal. There was also a use of AI for the image display on the LCD display, since the extent of the image conversion was a bit too much for us to do.

Acknowledgements